WEEK 2 · DAY 3 · INSTRUCTOR ONLY

Quiz 1 Answer Key

Weeks 1–2 · shuffled version

Instructor only. Built from the original `Time-series/MCQ/week-03-mcq.csv` (that file is unchanged), with 2 questions edited: see *Changes from the original* below. Options were reshuffled with a fixed seed (2603) so the correct letters are spread out: multiple-choice answers sit at A ×6, B ×7, C ×7, D ×6 (in the original, most answers were option B). The 4 True/False questions keep their original order.

#	Answer	Correct option	What it tests	Level
1	**C**	The order of the rows carries information	What makes data a time series	beginner
2	**D**	Seasonality	The four components	beginner
3	**B**	False	Noise versus change	beginner
4	**A**	A cycle, because the length is not fixed in advance	Seasonality versus cycle	intermediate
5	**B**	To make missing days appear as NaN instead of silently disappearing	Declaring the frequency	intermediate
6	**D**	Convert both to the same time zone first	Mixed time zones	intermediate
7	**A**	For bounded measures that live near the top, with the range stated on the chart	Honest y-axis	intermediate
8	**B**	The daily minimum	Mean hides the outage	beginner
9	**A**	Small multiples on a shared y-axis	Comparing six regions	beginner
10	**C**	Which region fell furthest from its own starting point	Indexing to a baseline	intermediate
11	**A**	True	Chart titles	beginner
12	**A**	Recommendation	Four-sentence finding	beginner
13	**B**	It contains every weekday exactly once, so the weekend effect cancels	Why seven days	beginner
14	**C**	It reacts late, roughly half a window behind the event	Trailing windows	intermediate
15	**B**	False	Centred windows	intermediate

#	Answer	Correct option	What it tests	Level
16	**B**	A seven-day mean genuinely does not exist until day seven	First rows are NaN	beginner
17	**A**	Follows the data closely and reacts fast	Exponential smoothing alpha	intermediate
18	**D**	Short outages vanish inside the average	Cost of smoothing	beginner
19	**A**	observed = trend + seasonal + residual	The decomposition equation	beginner
20	**D**	7	Choosing the period	intermediate
21	**C**	Multiplicative	Additive or multiplicative	intermediate
22	**C**	The residual	Where to investigate	beginner
23	**D**	Three	Minimum cycles	intermediate
24	**B**	Use STL with robust=True	Robust decomposition	advanced
25	**B**	The correlation of a series with a copy of itself shifted by one step	Definition	beginner
26	**C**	There is a weekly season with period 7	Reading the correlogram	beginner
27	**C**	A trend	Slow decay	intermediate
28	**A**	True	Lag 0	beginner
29	**B**	Neighbouring rows are near-copies, so the test set leaks into training	Random splits	intermediate
30	**D**	Mostly shared trend	Two trending series	advanced

Changes from the original

The same fixes are in the weekly question banks (`week-N/MCQ/`).

- **Q5** (reworded): It said "the code book", which participants never see.
- **Q30** (corrected): On the real amplifier file the correlation only drops from 0.59 to 0.40 after differencing, which is not a collapse. Replaced with the made-up suya example taught in W2D2 (0.97 to 0.06).

Marking it together (10 minutes)

1. Read out the letters; everyone marks their own paper.
2. Ask for hands up on each question. Any question that more than a third of the room got wrong, explain it again with the matching day's slide.

3. Map each topic back to its day: structure → W1D1 · charts → W1D2 · rolling windows → W1D3 · decomposition → W2D1 · autocorrelation → W2D2.

Quick topic tally

Topic	Questions
structure of time-series data	1, 2, 3, 4, 5, 6
visualising time-series and data storytelling	7, 8, 9, 10, 11, 12
rolling windows and smoothing	13, 14, 15, 16, 17, 18
decomposing an operational time series	19, 20, 21, 22, 23, 24
autocorrelation in operational data	25, 26, 27, 28, 29, 30